

UniMate AI

AI-Powered Smart University Platform

Project Documentation

Document No.: 04

Software Requirements Specification (SRS)

Part IV

External Interface Requirements

IEEE 29148 Inspired

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1. Introduction

This section defines the external interfaces required for UniMate AI to communicate with users, external systems, databases, and third-party services.

External interfaces describe how different components interact with each other and define the communication standards, data exchange methods, and user interaction requirements.

The main external interfaces of UniMate AI include:

- User Interface
- Mobile Application Interface
- Dashboard Interface
- Backend API Interface
- Database Interface
- Authentication Interface
- AI Service Interface
- Cloud Storage Interface

2. Interface Categories

The UniMate AI platform consists of the following external interfaces:

Interface	Description
User Interface	Interaction between users and applications
Mobile Interface	Android application interface
Dashboard Interface	Web-based administrative interface
API Interface	Communication between clients and backend
Database Interface	Backend communication with PostgreSQL
Authentication Interface	User identity verification
AI Interface	Communication with Gemini API
Storage Interface	Cloud file management

3. User Interface Requirements

UI-001: General Interface Design

Description:

The system shall provide a simple and consistent user interface across all platforms.

Requirements:

- Clear navigation.
- Consistent colors and typography.
- Logical screen organization.
- User-friendly interaction.

Actors: All Users | Priority: High

4. Android Application Interface Requirements

The Android application is designed for students and doctors.

Technology: Kotlin, Jetpack Compose, Material Design

UI-002: Authentication Screens

The application shall provide:

Login Screen — Components:

- Email Field
- Password Field
- Login Button
- Forgot Password Option

Registration Screen — Components:

- Name
- Email
- Password
- Role Selection

UI-003: Student Interface

The student application shall provide access to:

Home Screen — Contains:

- Available Courses
- Recent Activities
- Notifications

Course Screen — Displays:

- Course Information
- Lectures
- Assignments
- Files

AI Assistant Screen — Provides:

- Chat Interface
- Message History
- AI Responses

Profile Screen — Provides:

- Personal Information
- Settings
- Logout

UI-004: Doctor Interface

Doctors shall have access to:

Course Management — Functions:

- Create Course
- Edit Course
- Upload Materials

Assignment Management — Functions:

- Create Assignment
- View Submissions
- Provide Feedback

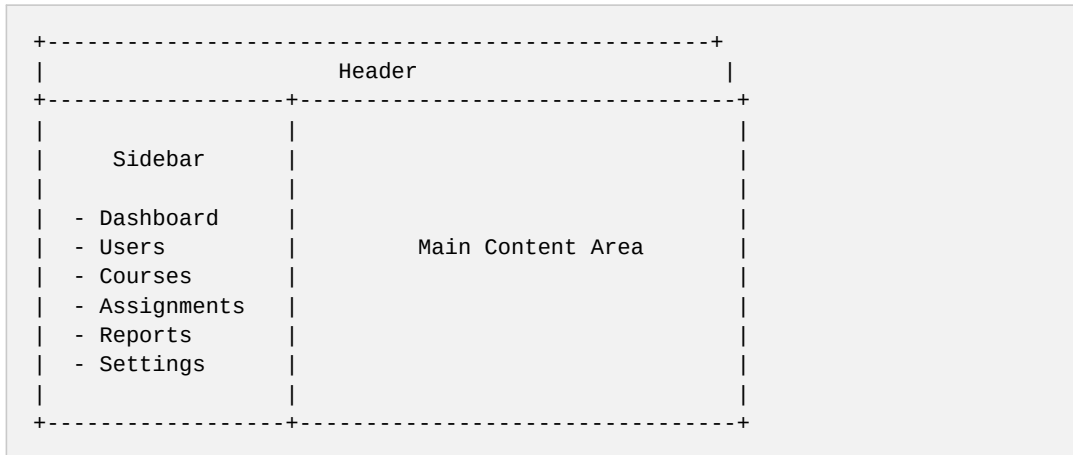
5. Dashboard Interface Requirements

The dashboard is a web application designed for administrators and doctors.

Technology: React, TypeScript, Material UI

UI-005: Dashboard Layout

The dashboard shall contain:



UI-006: Administrator Dashboard

The administrator interface shall provide:

User Management — Features:

- View Users
- Search Users
- Edit Users
- Disable Accounts

Reports — Displays:

- User Statistics
- Course Statistics
- System Activity

UI-007: Doctor Dashboard

The doctor dashboard shall provide:

- Course Management
- Assignment Management
- Student Monitoring
- File Management

6. Backend API Interface Requirements

The backend communicates with:

- Android Application
- React Dashboard

Communication Protocol: HTTPS + REST API

API-001: REST Communication

The system shall provide RESTful endpoints.

```
GET /api/courses
POST /api/auth/login
POST /api/files/upload
```

API-002: Data Format

All API communication shall use JSON.

```
{
  "success": true,
  "data": {}
}
```

API-003: Authentication Header

Protected endpoints require:

```
Authorization: Bearer JWT_TOKEN
```

API-004: Error Response Format

The API shall return standardized errors.

```
{
  "success": false,
  "message": "Unauthorized access"
}
```

7. Database Interface Requirements

Database Technology: PostgreSQL

DB-001: Database Communication

The backend shall communicate with PostgreSQL using:

- Entity Framework Core
- Database Providers

DB-002: Database Operations

The system shall support Create, Read, Update, Delete (CRUD Operations).

DB-003: Data Integrity

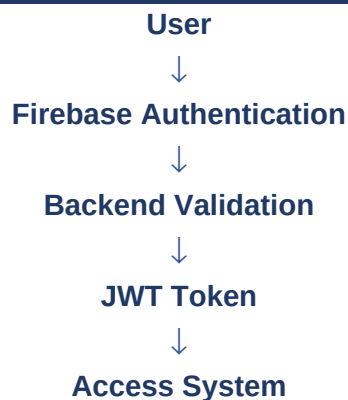
The database shall enforce:

- Primary Keys
- Foreign Keys
- Constraints
- Relationships

8. Authentication Interface Requirements

Authentication Technology: Firebase Authentication, JWT

AUTH-001: Authentication Flow



AUTH-002: Token Management

The system shall:

- Generate tokens.

- Validate tokens.
- Expire invalid sessions.

AUTH-003: Role Verification

The backend shall verify:

- User Identity.
- User Role.
- Access Permissions.

9. AI Service Interface Requirements

AI Provider: Gemini API

AI-001: AI Request Communication

The backend shall communicate with Gemini through secure API requests.

AI-002: AI Features

The interface shall support:

- Question Answering.
- Text Summarization.
- Academic Assistance.

AI-003: AI Response Handling

The system shall:

- Receive AI responses.
- Store conversation history.
- Display results to users.

10. Cloud Storage Interface Requirements

Storage Provider: Supabase Storage

STORAGE-001: File Upload Interface

The system shall support:

- Uploading files.
- Validating file types.
- Checking file size.

STORAGE-002: File Access Control

The system shall ensure:

- Private storage.
- Authorized access.
- Secure URLs.

STORAGE-003: File Management

Users shall be able to:

- Upload files.
- Download files.
- Delete files according to permissions.

11. Notification Interface Requirements

Notification Service — Future Integration: Firebase Cloud Messaging (FCM)

NOT-001: Notification Delivery

The system shall support notifications for:

- New assignments.
- Course updates.
- Important announcements.

12. Hardware Interface Requirements

The mobile application requires:

Minimum:

- Android device.
- Internet connection.
- Storage permission for files.

Optional:

- Camera (future features).
- Microphone (future AI voice features).

13. Communication Requirements

All communication between system components shall use:

Communication	Protocol
Android ↔ Backend	HTTPS REST
Dashboard ↔ Backend	HTTPS REST
Backend ↔ Database	Secure Database Connection
Backend ↔ AI	HTTPS API
Backend ↔ Storage	HTTPS API

14. Interface Security Requirements

All interfaces must implement:

- HTTPS encryption.
- Authentication validation.
- Authorization checks.
- Secure data transfer.
- Input validation.

15. Summary

Interface	Technology
Android Application	Kotlin + Jetpack Compose
Dashboard	React + TypeScript
Backend API	ASP.NET Core REST API
Database	PostgreSQL
Authentication	Firebase + JWT
AI	Gemini API
Storage	Supabase Storage

16. Conclusion

The External Interface Requirements define how UniMate AI components communicate with users, internal modules, and external services. By establishing clear interface standards, the system ensures reliable communication, secure data exchange, and easier future expansion.

End of Part IV

Document No. 04 – Software Requirements Specification (SRS) – Part IV: External Interface Requirements
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